Lab 9
Introduction to RC Circuits in the Time and Frequency-Domains

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I. Introduction

This lab had two main objectives. The first was to build a simple RC circuit and drive it with a square wave to watch the capacitor charge and discharge, then calculate the time constant τ from the decay on the oscilloscope. The second was to drive the same circuit with sine waves from 0.1 Hz to 10,000 Hz and compare the input and output in both magnitude and phase.

II. Results

1. Setting Up the RC Circuit

A 20 kΩ resistor was connected in series with a 0.1 μF capacitor, with the function generator driving the pair and the voltage across the capacitor taken as the output. The function generator was set to produce a square wave with a 1 V amplitude (2 Vpp) and a +1 V DC offset, so the input switched between 0 V and 2 V.

2. Choosing the Square Wave Frequency

The expected time constant of the circuit is:

$$\tau = RC = (20 \times 10^3\ \Omega)(0.1 \times 10^{-6}\ \text{F}) = 2\ \text{ms}$$

To see the capacitor fully charge and discharge, each half of the square wave should last at least $10\tau = 20$ ms. This gives a period and frequency of:

$$T = 2(20\ \text{ms}) = 40\ \text{ms}$$
$$f = 1/T = 1/0.040\ \text{s} = 25\ \text{Hz}$$

The oscilloscope confirmed a 25.000 Hz square wave with a peak-to-peak voltage of 2.07 V (Fig. 1).

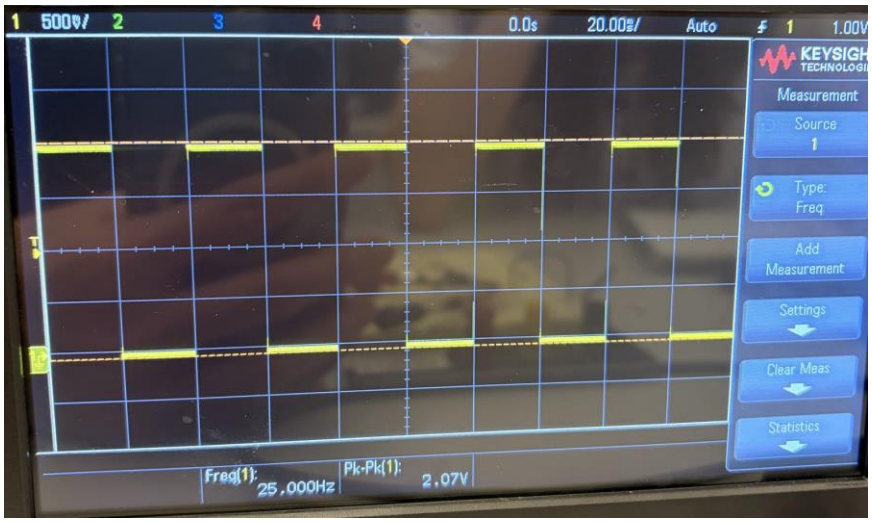


Figure 1: 25 Hz, 2 Vpp square wave input with a +1 V DC offset.

3. Observing the Capacitor Voltage

The input was displayed on channel 1 and the capacitor voltage on channel 2, both at 1.00 V/div. The time axis was then set to 1.000 ms/div to zoom into one falling edge of the square wave. As

shown in Fig. 2, when the input dropped, the capacitor voltage did not drop with it, but instead decayed smoothly toward 0 V over several milliseconds.

4. Measuring the Time Constant

The cursors were placed on the capacitor voltage at the start of the decay and at a later point on the curve (Fig. 2), which gave:

$t_1 = 20.00$ ms, $V_1 = 1.935$ V

$t_2 = 21.56$ ms, $V_2 = 0.912$ V

Since $V_c(t) = V_0 e^{-t/\tau}$, dividing the voltages at the two times cancels V_0 and leaves $V_1/V_2 = e^{\Delta t/\tau}$, so:

$$\tau = \Delta t / \ln(V_1/V_2) = 1.56 \text{ ms} / \ln(1.935/0.912) = 1.56 \text{ ms} / 0.752 \approx 2.07 \text{ ms}$$

This is about 4% higher than the expected 2 ms, so it is consistent with the circuit and with what was discussed in class. The small difference likely comes from the tolerances of the resistor and capacitor, whose real values can be several percent off from their labels, along with small errors in placing the cursors on the trace.



Figure 2: Capacitor voltage decay (channel 2) at 1.000 ms/div, with cursors at 20.00 ms and 21.56 ms.

5. Sine Wave Input at 80 Hz

The function generator was switched to an 80 Hz sine wave with a 1 V amplitude, still set to high Z, and the input measured 2.05 Vpp on the oscilloscope. The capacitor voltage was measured on channel 2 (Fig. 3).

In magnitude, the output was smaller than the input. The output measured about 1.41 Vpp (the reading moved between about 1.39 and 1.41 V), compared with 2.05 Vpp at the input, so:

$V_{out}/V_{in} = 1.41/2.05 \approx 0.69$

In phase, the output lagged the input, with each of its peaks arriving after the matching input peak. The oscilloscope measured a phase difference of 45.5°, or about π/4 radians, so the output was about 1/8 of a cycle behind the input.

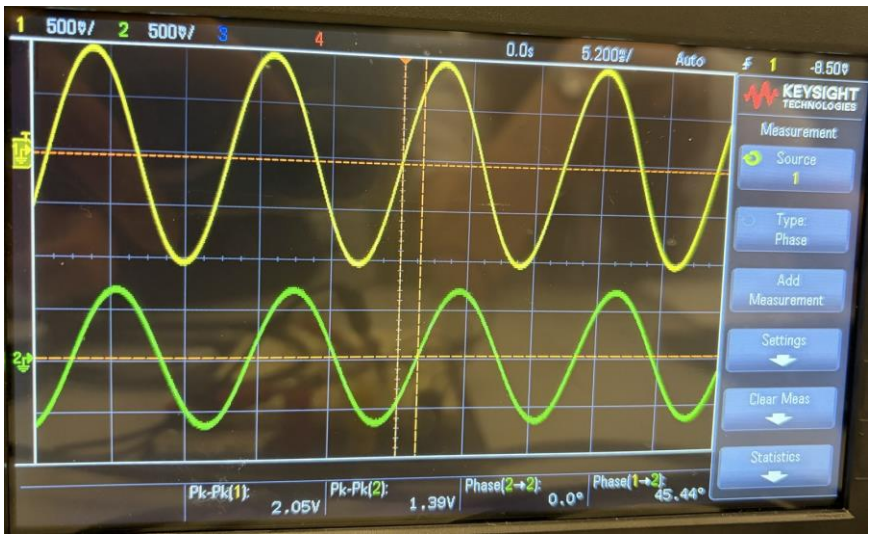


Figure 3: 80 Hz sine wave input (channel 1) and capacitor voltage output (channel 2).

6. Frequency Response

With the input kept at the same amplitude, the frequency was raised to 100, 1000, and 10,000 Hz, then lowered to 10, 1, and 0.1 Hz. The results are shown in Table 1.

Table 1: Input and output of the RC circuit from 0.1 Hz to 10,000 Hz.

Frequency (Hz)	V_{in} (Vpp)	V_{out} (Vpp)	V_{out}/V_{in}	Phase (°)
0.1	0.445	0.436	0.98	0.3
1	1.989	1.949	0.98	0.9
10	2.05	1.97	0.96	9.04
80	2.05	1.41	0.69	45.5
100	2.05	1.25	0.61	51.5
1000	2.05	0.22	0.11	91.4
10,000	2.05	0.035	0.017	96.2

As the frequency went up, the output got smaller and lagged further behind the input. From 100 Hz to 10,000 Hz, the output fell from 61% of the input to under 2%, and the phase grew from 51.5° to about 90°. As the frequency went down, the output got closer to the input in both magnitude and phase, and at 1 Hz and 0.1 Hz it was 98% of the input with less than 1° of phase difference. This means the circuit passes low frequencies and blocks high ones, so it acts as a low-pass filter. At low frequencies, the input changes slowly enough for the capacitor to charge and follow it. At high frequencies, the input changes before the capacitor has time to charge, so its voltage stays small and falls behind.

Two results did not fit this pattern exactly. At 1000 Hz and 10,000 Hz, the phase read 91.4° and 96.2°, slightly over 90°. A single RC circuit can only shift the phase by up to 90°, so the extra was likely measurement error, since the output was only 0.22 V and 0.035 V and noise made the phase measurement less accurate. Also, at 0.1 Hz the input read only 0.445 Vpp instead of about 2 V. This was likely because one cycle at 0.1 Hz lasts 10 s, so the oscilloscope did not capture a full cycle and measured only part of the swing. Since both channels were measured the same way, the ratio of 0.98 still matched the 1 Hz result.

III. Discussion and Conclusions

This lab showed how a capacitor responds to changes in its input in both the time and frequency domains. With a 25 Hz square wave, the capacitor voltage decayed exponentially instead of switching instantly, and the time constant calculated from the decay was about 2.07 ms, within about 4% of the expected RC = 2 ms.

The sine wave tests showed that the same circuit behaves differently depending on frequency. At low frequencies, the output followed the input almost exactly, while at high frequencies, the output became very small and lagged the input by about 90°. The two parts of the lab are connected by τ. The 80 Hz test frequency is close to $1/(2\pi\tau) \approx 80$ Hz, where an RC circuit's output is expected to drop to about 0.707 of the input with a 45° lag, which matches the measured 0.69 and 45.5°. Overall, this lab connected RC circuits from class to real measurements, thus demonstrating two ideas: a capacitor cannot change its voltage instantly, so it charges and discharges with a time constant of $\tau = RC$, and because of this, an RC circuit passes low frequencies and blocks high ones.

IV. References

[1] Dr. Iman Salama. “Lab 9 – Introduction to RC Circuits in the Time and Frequency-Domains.” Northeastern University. 23 August 2021.